

Symbol Mountain

Production Freeze Checklist

Scenes 01-04: Complete Production Readiness Verification

Last Updated: 2026-04-17

Status: Ready for Comprehensive Freeze Review

Quick Navigation

- 1. Master Gate (All Scenes)
- 2. Scene 01 - Modak
- 3. Scene 02 - Pond
- 4. Scene 03 - Symbol
- 5. Scene 04 - Sacred Assembly (Detailed)
- 6. Cross-Scene Features
- 7. Final Sign-Off

Master Gate: All 4 Scenes Must Pass These Core Requirements

MG-1: Shared Chrome (T01-T10, T13, T18-T22)

Opening Modal (T01)

- ☐ Imported from correct shared path (`../../shared/components/OpeningModal`)
- ☐ Rendered on scene mount (first phase)
- ☐ Baloo 2 title font — no system fonts
- ☐ Nunito description text — correct weight
- ☐ Action button is ≥ 60 px touch target
- ☐ CTA explains mechanic clearly in age 5-12 language
- ☐ Does NOT re-trigger on tab return

- ☐ VO narration syncs to modal text
- ☐ Modal closes cleanly on CTA tap — no lingering overlay
- ☐ Correct zone color scheme applied (Symbol Mountain = #FF5722 primary)

Completion Celebration (T02)

- ☐ SceneCompletionCelebration imported from correct path
- ☐ Fires exactly on win condition (no early/late trigger)
- ☐ Title and subtitle displayed correctly
- ☐ Affirmation line present and culturally grounded
- ☐ "Continue" button advances to next scene
- ☐ "Replay" button fully resets scene state (localStorage cleared)
- ☐ Confetti/sparkles fire ONCE only — no loop
- ☐ Ganesha presence visible (pose=blessing or appropriate)
- ☐ Modal closes cleanly after button tap — no stuck overlay
- ☐ ProgressManager.updateProgress() called on completion

Audio Toggle (T04)

- ☐ AudioToggle component renders (usually top-right UI area)
- ☐ **Toggle OFF stops ALL VO lines immediately with no stutter**
- ☐ **Toggle OFF stops ALL SFX (tap, chime, sparkle) immediately**
- ☐ Toggle ON resumes VO queue without replay or skip
- ☐ Audio preference persists across tab switch via localStorage
- ☐ Audio preference persists across scene reload
- ☐ Audio preference persists across all 4 Symbol Mountain scenes
- ☐ Visual indicator clear (toggle style changes visually)
- ☐ Muted state indicator visible to child users

Home Button (T05)

- ☐ HomeButton visible in consistent position (usually top-left)
- ☐ Mid-scene tap opens confirmation dialog
- ☐ "Cancel" returns to scene with no state change

- ☐ "Home" calls ProgressManager.updateProgress() before navigation
- ☐ Navigates safely to home/map screen
- ☐ Button is ≥60px touch target

Zone Badge (T13)

- ☐ ZoneBadgeButton displays correct zone label (Symbol Mountain)
- ☐ Zone color scheme applied (#FF5722 primary, #FFD700 accent)
- ☐ Badge updates progress bar (1/4 → 2/4 → 3/4 → 4/4)
- ☐ Badge position consistent across all 4 scenes
- ☐ Shows completion checkmark (✓) after zone fully complete

Voice Guidance (T07 + T09)

- ☐ useVoiceGuidance hook integrated in main component
- ☐ idleTimeout configured (typically 15-20s)
- ☐ resumeDelay set (typically 3000ms)
- ☐ enableMusic set appropriately
- ☐ Return hint VO plays after tab return (after resumeDelay)
- ☐ Idle hint VO plays on inactivity
- ☐ stopVoice() called on scene unmount
- ☐ No stale voice refs after tab switch

Tab Visibility + Pause (T08)

- ☐ usePauseAwareTimeout hook integrated
- ☐ onHide: stops all VO (stopVoice + stopSpokenVoice)
- ☐ onHide: clears idle hint timers
- ☐ onHide: pauses celebration if showing
- ☐ onShow: resumes celebration
- ☐ onShow: triggers return hint VO after resumeDelay
- ☐ No duplicate timers after tab return
- ☐ No stuck animations (spinning, pulsing) after return

Gesture Mapping (T18)

- ☐ Ganesha gesture defined per scene phase
- ☐ Blessing pose on completion
- ☐ Pointing/nudging pose during idle hints
- ☐ Wave or celebrate pose on symbol reveal
- ☐ Gesture transitions smoothly (no stutter)
- ☐ Gesture timing aligns with VO/sparkle moments

Image Audit (T19)

- ☐ Zero 404s in DevTools Network tab
- ☐ Background image correct per scene (no wrong fallback)
- ☐ All symbol icon variants load (colored + gray where applicable)
- ☐ Popup/card images match per-symbol content
- ☐ Side rail icons load correctly
- ☐ Completion modal badge image loads
- ☐ No missing PNG/SVG in any phase

Responsive Design (T20)

- ☐ Desktop (1280×800): all text readable, no overflow
- ☐ Tablet (768×1024): touch targets all ≥60px, layout adapts
- ☐ Mobile (375×812): cards/modals fit viewport, scrollable if needed
- ☐ clamp() functions used for responsive scaling (no hardcoded px)
- ☐ Font scaling tested at all breakpoints
- ☐ Images scale proportionally (no stretching/squishing)

Test Cases (T21)

- ☐ Test matrix documented in TESTCASES.md
- ☐ Edge cases logged (rapid tap, tab switch mid-phase, reload timing)
- ☐ All critical paths tested (early phase, mid phase, completion)
- ☐ Cross-browser tested (Chrome, Safari, Firefox if applicable)

Idle Hints (T22)

- ☐ Idle hint VO triggers after configured delay (15-20s typically)
- ☐ Hint targets correct interactive element per phase
- ☐ Pointer emoji or glow animation visible
- ☐ Hint clears on any user interaction
- ☐ Hint clears on phase transition
- ☐ Hint does NOT re-trigger after cleared in same phase
- ☐ Reload mid-hint clears hint state safely

MG-2: State Persistence & Resume

Reload Behavior (All Scenes)

- ☐ Reload in `intro` phase → opening modal shows fresh (no resume popup)
- ☐ Reload in active gameplay phase → safe state restored (no stuck card/modal)
- ☐ Resume countdown displays (typically 3s) if resuming mid-scene
- ☐ Resume countdown does NOT block gameplay
- ☐ All transient UI states cleared on reload (modals, popovers, overlays)
- ☐ Completion state persists (if already won, don't restart)
- ☐ Symbol unlocks persist (side rail shows correct state)
- ☐ localStorage keys correctly namespaced (no cross-scene collision)

Tab Switch Behavior (All Scenes)

- ☐ Tab switch away: VO stops, timers pause, celebration pauses
- ☐ Tab switch back: completion resumes visually (no jank)
- ☐ Tab switch back: return hint VO plays after resumeDelay
- ☐ No duplicate VO lines after tab return (stale ref cleanup)
- ☐ No duplicate sparkle/animation timers after return
- ☐ Mid-modal tab switch: modal still visible on return
- ☐ Mid-drag/interaction tab switch: can resume safely (no hung state)

Scene 01 - Modak (NewModakSceneV7)

Phase Sequence:

```
intro
  → mooshika-search (find Mooshika in scene)
  → modak-collect (collect modaks into basket)
  → modak-feed (drag modaks from basket to rock)
  → belly-reveal (Ganesha belly shows Modak symbol)
  → symbol-card (flip card shows symbol, affirmation, meaning)
  → completion
```

S01-A: Visual Assets & Layout

Background & Main Assets

- ☐ Background image (scene1-background.jpg or similar) loads with no 404
- ☐ Background fills viewport without tiling or stretching
- ☐ Mooshika sprite renders at correct size
- ☐ Modak sprites (multiple instances) render crisply
- ☐ Basket asset appears on-screen when active
- ☐ Rock asset positioned correctly
- ☐ Ganesha belly area visually aligns with reveal moment
- ☐ All assets use Symbol Mountain color scheme (#FF5722 accents)

Symbol Card UI

- ☐ Card appears with smooth animation (scale/fade)
- ☐ Card front shows symbol icon (Modak)
- ☐ Card back shows "Modak" text + association ("Ganesha's favorite sweet")
- ☐ Card affirmation line ("I enjoy sweet moments") renders
- ☐ Card close/dismiss button is ≥60px
- ☐ Card flies to sidebar smoothly on completion (no jump)
- ☐ Card lands in sidebar icon position

- ☐ Sidebar icon highlights when new symbol unlocked

Sparkle/Fireworks

- ☐ Sparkles appear on modak tap (collection)
- ☐ Sparkles appear on correct drop (basket → rock)
- ☐ Fireworks trigger on symbol reveal (belly)
- ☐ Fireworks trigger on card completion (move to sidebar)
- ☐ Sparkles/fireworks render ABOVE all game elements (z-index)
- ☐ No clipping at viewport edges
- ☐ Animation duration is satisfying (not too fast/slow)

S01-B: Game Mechanics

Mooshika Discovery

- ☐ Mooshika tap (or click) registers as interaction
- ☐ Discovery triggers phase transition to modak-collect
- ☐ Discovery VO plays ("I love modaks...")
- ☐ Visual feedback (glow, bounce) on Mooshika

Modak Collection

- ☐ Modak tap increments collection counter
- ☐ Counter updates visually (1/5, 2/5, etc. or similar)
- ☐ Basket visual updates (modaks appear in basket)
- ☐ Collection continues until threshold reached
- ☐ Reaching threshold auto-advances to modak-feed phase

Modak Feed Mechanic

- ☐ Drag from basket to rock hitbox detects correctly
- ☐ Correct drop triggers sparkle + celebration
- ☐ Wrong drop (off target) returns modak to basket with gentle feedback
- ☐ Drop counter increments per successful feed

- ☐ Feed threshold triggers belly reveal phase

Belly Reveal

- ☐ Ganesha belly visually changes (opens, glows, reveals symbol)
- ☐ Symbol icon (Modak) displays on belly
- ☐ Completion VO plays ("Look! The Modak symbol...")
- ☐ Transition to symbol-card phase is smooth

Symbol Card Flow

- ☐ Card appears with flip animation
- ☐ Card content (name, meaning, affirmation) displays correctly
- ☐ Card can be flipped between front/back (if interaction enabled)
- ☐ Card tap-to-fly works (flies to sidebar)
- ☐ Sidebar icon locked/unlocked state updates
- ☐ Completion modal triggers after card settles

S01-C & S01-D: Voice, Hints, Reload & Resume

VO Triggers

- ☐ Intro VO ("Modak is my favorite sweet...") plays once on scene enter
- ☐ Mooshika discovery VO plays once ("I love modaks...")
- ☐ Collection phase encouragement VO plays ("Collect more modaks...")
- ☐ Feed phase instruction VO plays ("Feed me the modaks...")
- ☐ Belly reveal celebration VO plays ("The Modak symbol is unlocked...")
- ☐ Completion VO plays on card completion
- ☐ No VO lines duplicate after tab switch/reload
- ☐ All VO uses age 5-12 friendly language

Idle Hints & Tab Switch

- ☐ Idle hint triggers after 15s+ inactivity in active phase
- ☐ Hint targets correct object (Mooshika, modak, basket, rock per phase)

- ☐ Tab return: return hint VO plays after 3s resume delay
- ☐ Return hint contextual ("You were collecting modaks...")
- ☐ Hint does NOT re-trigger mid-phase

Reload Scenarios

- ☐ Reload in mooshika-search → Mooshika still findable, phase restarts
- ☐ Reload in modak-collect (5/5 collected) → counter preserved, ready to feed
- ☐ Reload in modak-feed (2/3 fed) → fed count preserved, continue feeding
- ☐ Reload in belly-reveal → reveals still showing, card ready
- ☐ Reload after completion → completion state persists (no restart)
- ☐ Resume countdown shows "Resuming in 3..." for active phases

Scene 02 - Pond (PondSceneSimplifiedV4)

Phase Sequence:

```
intro
  → lotus-waiting (lotus flowers wait to bloom)
  → lotus-bloom (tap to bloom each lotus)
  → golden-lotus (optional: special golden lotus interaction)
  → trunk-reveal (elephant trunk reveals symbol)
  → symbol-card (flip card reveals symbol meaning)
  → completion
```

S02-A: Visual Assets & Layout

Background & Main Assets

- ☐ Background image (pond scene) loads with no 404
- ☐ Lotus flowers render at correct positions (grid or scattered)
- ☐ Lotus closed/bloomed states both visible and animate smoothly
- ☐ Golden lotus (if used) visually distinct from regular lotus
- ☐ Elephant/trunk asset positioned correctly

- ☐ Water/ripple effects visible (if animated)
- ☐ All colors match Calm theme (soft greens, blues, Symbol Mountain accents)

Lotus Bloom Animation

- ☐ Lotus closed state (bud, folded petals)
- ☐ Lotus bloom state (open petals, golden center)
- ☐ Transition animation smooth (0.5-1s duration)
- ☐ Bloom sparkle/glow effect visible
- ☐ Audio feedback (soft chime or water sound) on bloom

Trunk Reveal & Symbol Card

- ☐ Trunk emerges from water smoothly
- ☐ Symbol card appears after trunk reveal
- ☐ Card visuals correct (Lotus symbol, meaning, affirmation)
- ☐ Card flies to sidebar on completion
- ☐ Sidebar icon updates (Lotus unlocked)

S02-B: Game Mechanics

Lotus Tap Mechanics

- ☐ Lotus tap registers reliably (hitbox correct)
- ☐ Tap triggers bloom animation
- ☐ Already-bloomed lotus cannot be re-tapped (locked state)
- ☐ Bloom counter increments
- ☐ Visual feedback (sparkle/glow) on tap

Progression Requirements

- ☐ All lotus blooms required before trunk reveal
- ☐ Lotus order does NOT matter (can bloom any lotus)
- ☐ Golden lotus (if any) counted toward threshold
- ☐ Reaching threshold auto-triggers trunk reveal

Trunk Reveal & Completion

- ☐ Trunk emerges after threshold reached
- ☐ Trunk reveal VO plays ("The trunk appears...")
- ☐ Symbol card appears (auto or on trunk tap)
- ☐ Card displays Lotus symbol + meaning + affirmation
- ☐ Card move to sidebar triggers completion modal

S02-C & S02-D: Voice, Hints, Reload & Resume

VO Triggers

- ☐ Intro VO ("This pond is calm and beautiful...")
- ☐ Bloom encouragement VO ("Tap the lotus to make it bloom...")
- ☐ All-bloomed transition VO ("The lotus is so beautiful...")
- ☐ Trunk reveal VO ("Look! The trunk brings the symbol...")
- ☐ Completion VO on card move to sidebar

Idle Hints & Reload

- ☐ Idle hint triggers after 15s+ in bloom phase
- ☐ Reload in intro → opening modal shows fresh
- ☐ Reload in lotus-bloom (3/5 bloomed) → bloom count preserved, continue
- ☐ Reload after trunk-reveal → trunk still visible
- ☐ Reload after completion → completion state persists

Scene 03 - Symbol (SymbolMountainSceneV3)

Phase Sequence:

```
intro
  → eyes-game (telescope/sight interaction)
  → ears-game (rhythm/listening interaction)
```

- tusk-game (assembly/construction interaction)
- all-complete (all 3 sub-games done)
- completion

S03-A: Visual Assets & Layout

- ☐ **Eyes Game:** Telescope/viewing asset renders, symbol preview visible through telescope, focus/zoom animation
- ☐ **Ears Game:** Musical notes/instruments render, rhythm pattern visual (if shown) correct, audio visualizer or animation
- ☐ **Tusk Game:** Assembly puzzle pieces visible and draggable, tusk shape updates as pieces placed, completed tusk renders with glow
- ☐ **Progress & Sidebar:** Progress indicator (3 sub-games) visible, sidebar shows symbols as unlocked, icon states update

S03-B: Game Mechanics

Eyes, Ears, Tusk Games

- ☐ Eyes: Tap/click telescope → opens/zooms → eyes symbol preview shown → unlocks and card appears
- ☐ Ears: Musical notes play → child interaction (tap/hold notes) → celebration → unlocks and card appears
- ☐ Tusk: Puzzle pieces visible and draggable → correct placement locks piece → progress bar updates → all pieces placed triggers celebration

All-Complete Transition

- ☐ After all 3 games done, final celebration triggers
- ☐ Fireworks appear
- ☐ Completion modal appears
- ☐ All 3 symbols visible in sidebar

S03-C & S03-D: Voice, Hints, Reload & Resume

VO & Idle Hints

- ☐ Eyes game entry VO ("Look through the telescope...")
- ☐ Ears game entry VO ("Listen to the rhythm...")

- ☐ Tusk game entry VO ("Assemble the tusk...")
- ☐ Idle hints contextual to sub-game phase (glow/pointer)

Reload Safety

- ☐ Reload in eyes-game (unlocked) → eyes still unlocked, eyes card gone
- ☐ Reload in ears-game (mid-rhythm) → rhythm phase resets (safe)
- ☐ Reload in tusk-game (1/3 pieces) → piece count preserved
- ☐ Reload after all-complete → all 3 symbols persist in sidebar

Scene 04 - Sacred Assembly (SacredAssemblySceneV8) [MOST COMPLEX]

Phase Sequence:

```
intro
  → round-start (card appears, lands on side)
  → card-play (card flips, guide arrow appears, child taps symbol)
  → round-feedback (celebration or retry)
  → [repeat for all 8 symbols]
  → round-8-complete (all symbols placed)
  → completion
```

 **CRITICAL SCENE:** This is the most complex Symbol Mountain scene. All mechanics must be rock-solid for production.

S04-A: Visual Layout & Assembly Canvas

Sacred Background & Canvas

- ☐ Sacred Assembly background loads with no 404
- ☐ Ganesha silhouette/base rendered correctly
- ☐ All 8 symbol drop zones (head, neck, arms, torso, legs) visible
- ☐ Drop zones have subtle visual guides (color, glow, or outline)
- ☐ Drop zone hitboxes are generous (easy to hit on mobile)

☐ Progress bar/counter visible ("1/8", "2/8", etc.)

☐ Symbol count updates as each is placed

Card Position & Movement

☐ **Appearance:** Card appears with entrance animation (scale + fade), lands in card-slot position (left side, typically)

☐ **Flipped State:** Card flips smoothly (left to right rotation), content visible: symbol name + association text, guide arrow appears from card toward Ganesha

☐ **Side Movement:** After correct placement, card slides LEFT off-screen smoothly (duration ~0.5s), off-screen card does NOT linger, new card appears on-screen

Card Content Mapping (CRITICAL)

☐ Card 1: Head symbol + "The head helps me think"

☐ Card 2: Neck symbol + "The neck connects..."

☐ Card 3: Right arm symbol + "The right arm..."

☐ Card 4: Left arm symbol + "The left arm..."

☐ Card 5: Torso symbol + "The belly holds..."

☐ Card 6: Right leg symbol + "The right leg..."

☐ Card 7: Left leg symbol + "The left leg..."

☐ Card 8: Final symbol + "All together, we are whole"

☐ **ALL TEXT IS CORRECT (not swapped or wrong symbol)**

Sidebar & Symbol Display

☐ Sidebar shows all 8 symbol icons in a vertical rail

☐ Icons display correctly (correct color, not stretched)

☐ As symbols are placed, sidebar icons light up/glow

☐ Locked symbols (not yet placed) have muted appearance

☐ Unlocked symbols have full color + glow

S04-C: Symbol Tap/Placement Mechanics (CRITICAL)

Symbol Selection & Zone Detection

☐ Symbol icon (on card or separate UI) is tappable (≥60px target)

- ☐ Tap registers correctly (no double-tap needed)
- ☐ Visual feedback on tap (glow, scale, highlight)
- ☐ Tapping wrong symbol (not current card symbol) does nothing
- ☐ Tapping correct zone places symbol
- ☐ Correct zone visual updates (symbol appears in zone, zone locks)
- ☐ Sparkle/celebration plays on correct placement
- ☐ Round advances immediately after correct placement

Wrong Zone & Already-Placed Protection

- ☐ Tapping wrong zone gives gentle feedback (shake, sound, or brief red flash)
- ☐ Wrong zone does NOT place symbol or lock
- ☐ After wrong tap, child can retry immediately (no delay)
- ☐ Tapping zone with already-placed symbol does nothing
- ☐ Zone is visually locked (appears grayed out or with checkmark)

S04-D & S04-E: Round Flow & Voice

Round Sequence

- ☐ Card appears (round N of 8)
- ☐ Card shows correct symbol + association text
- ☐ Guide arrow appears
- ☐ Child taps correct zone
- ☐ Placement triggers celebration (sparkle + brief VO)
- ☐ Card slides off-screen
- ☐ New card appears (round N+1)
- ☐ Pattern repeats through round 8

Round 8 & Completion

- ☐ After 8th correct placement, special celebration triggers (larger fireworks)
- ☐ Completion modal appears showing "All 8 symbols placed!"

- ☐ Progress bar shows 8/8
- ☐ All symbol zones show placed symbols
- ☐ ProgressManager.updateProgress() called

VO Triggers

- ☐ Opening modal VO plays once ("Welcome to the Sacred Assembly...")
- ☐ Intro VO before first round ("Let's place each symbol...") plays once per session
- ☐ Card VO plays as card flips (not before)
- ☐ VO timing: does NOT exceed card-play phase
- ☐ Idle hint VO triggers after 15s+ inactivity (once per round)
- ☐ Return hint VO contextual to current round on tab return

S04-F & S04-G: Idle Hints, Gestures, Reload & Tab Switch

Visual Idle Hints & Ganesha Gestures

- ☐ Idle level 1 (after 5s): Ganesha or current zone glows/pulses
- ☐ Idle level 2 (after 10s): Pointer emoji appears near target zone
- ☐ Idle level 3 (after 15s): Guide arrow thickens/brightens as nudge
- ☐ All hints clear on interaction (symbol tap or zone tap)
- ☐ Ganesha gestures: waves (round start) → points (play) → celebrates (correct placement) → shakes gently (wrong placement)

Reload Scenarios (CRITICAL)

- ☐ Reload in Round 1 (Card Appearing): Card re-appears and animates, round counter shows "1/8"
- ☐ Reload in Round 3 (Mid-Play): Round counter shows "3/8", zones 1-2 show placed symbols, Card 3 appears (flipped, arrow visible)
- ☐ Reload in Round 5 (Card Sliding): Round counter shows "5/8", zones 1-4 show placed symbols, card completes slide or restarts (graceful recovery)
- ☐ Reload After Round 8: Modal still visible, all 8 symbols placed and visible, all buttons functional

Tab Switch Scenarios

- ☐ Tab away: VO stops, timers pause, card frozen

- ☐ Tab back (3s later): Completion resumes visually, return hint VO plays, guide arrow re-appears
- ☐ Tab switch during celebration: Sparkle pauses, card slide resumes smoothly, no duplication
- ☐ Tab switch in Round 8: Fireworks resume, completion modal appears, safe navigation

Edge Cases

- ☐ Double-tap symbol: Recognized as single tap (no duplicate placement)
- ☐ Tap multiple zones in succession: Only first correct tap registers
- ☐ Rapid tapping: Only first tap counts (symbol "locks" after tap)
- ☐ Modal interruptions: Home button opens confirmation, "Cancel" resumes safely

Cross-Scene Features

Symbol Sidebar Popup System

Sidebar Icon Behavior

- ☐ Sidebar displays all 8 symbols (vertical rail, right or left side)
- ☐ Locked symbols appear muted/grayed out
- ☐ Unlocked symbols appear bright + glow
- ☐ Newly unlocked symbols highlight/pulse briefly
- ☐ Icon size consistent across all scenes

Popup Interaction

- ☐ Tap unlocked symbol icon → popup overlay appears
- ☐ Popup shows symbol name (e.g., "Modak")
- ☐ Popup shows symbol meaning/association (e.g., "Symbol of sweetness and devotion")
- ☐ Popup shows "Close" button (clear, ≥60px)
- ☐ Popup does NOT pause current phase (child can close and continue)
- ☐ Popup z-index correct (above gameplay, below alerts)

Popup Content Accuracy (ALL 8 SYMBOLS)

- ☐ Symbol 1: Modak — Correct image + meaning
- ☐ Symbol 2: Lotus — Correct image + meaning
- ☐ Symbol 3: Eyes — Correct image + meaning
- ☐ Symbol 4: Ears — Correct image + meaning
- ☐ Symbol 5: Tusk — Correct image + meaning
- ☐ Symbol 6: Trunk — Correct image + meaning
- ☐ Symbol 7: Mouse (Mooshika) — Correct image + meaning
- ☐ Symbol 8: Pot/Modak (variant) — Correct image + meaning
- ☐ **All meanings culturally accurate + age-appropriate**

Sidebar Persistence

- ☐ Unlocked state persists across scenes
- ☐ Unlocked state persists across reload
- ☐ Symbol progression order respected (Scene 1 unlocks Modak, etc.)

☒ Final Sign-Off

FS-1: Complete Test Matrix

Desktop Testing (1280×800)

- ☐ Scene 01 (Modak): Full playthrough, all phases
- ☐ Scene 02 (Pond): Full playthrough, all lotus bloomed
- ☐ Scene 03 (Symbol): All 3 sub-games completed
- ☐ Scene 04 (Sacred Assembly): All 8 rounds completed
- ☐ All 4 scenes with Audio ON
- ☐ All 4 scenes with Audio OFF
- ☐ Tab switch tested in early/mid/late phase of each scene
- ☐ Mid-scene reload tested at least 2x per scene (different phases)
- ☐ DevTools Console: Zero blocking errors or red logs

- ☐ DevTools Network: Zero 404 assets, all images load

Mobile Testing (375×812)

- ☐ Scene 01-04: Layout responsive (no overflow, readable text)
- ☐ All tap targets ≥60px on mobile (test on actual device if possible)
- ☐ Touch interactions responsive (no lag > 100ms)
- ☐ Card/popup positioning correct on small screen
- ☐ Font scaling appropriate (clamp()) functions working

Tablet Testing (768×1024)

- ☐ Intermediate viewport testing (landscape + portrait)
- ☐ Touch targets sized appropriately
- ☐ Sidebar and main UI layout adapt smoothly

Cross-Browser Testing

- ☐ Chrome (latest): Smooth, no jank
- ☐ Safari (iOS + macOS): Smooth, no jank
- ☐ Firefox (if applicable): Smooth, no jank

Audio & VO Verification

- ☐ All VO files (.mp3) load correctly (no 404)
- ☐ VO timing syncs with animations
- ☐ All SFX (tap, sparkle, chime) trigger at correct moments
- ☐ Audio toggle mutes/unmutes all sounds
- ☐ Audio preference persists across reload

FS-2: Production Readiness Checklist

Code Quality

- ☐ No TypeScript errors in any scene file
- ☐ No eslint warnings that block deployment

- ☐ All imports reference correct shared paths
- ☐ LocalStorage keys properly namespaced

Performance

- ☐ Scene load time < 2 seconds (on 3G)
- ☐ Animations smooth (60 FPS target, no dropped frames)
- ☐ No memory leaks on tab switch or reload (DevTools Memory check)
- ☐ Asset files optimized (images compressed, no oversized PNG)

Documentation

- ☐ Scene phase diagram documented (ascii or markdown)
- ☐ VO mapping documented (which VO line plays at which phase)
- ☐ localStorage key schema documented
- ☐ Asset paths documented (background, symbols, popups)

Compliance

- ☐ No copyrighted content (all assets proprietary or licensed)
- ☐ All Sanskrit text reviewed for cultural accuracy
- ☐ All content reviewed for age-appropriateness (5-12 years)

FS-3: QA & Production Sign-Off

Role	Name	Date	Status
QA Tester	_____	_____	PENDING
Senior UI/UX Reviewer	_____	_____	PENDING
Scene Owner / Madhurima	_____	_____	PENDING

FS-4: Deployment Checklist

- ☐ All critical blockers (Red items) resolved
- ☐ All test cases passed

- ☐ All 3 sign-offs obtained
- ☐ Release notes prepared (if applicable)
- ☐ Rollback plan documented (if applicable)
- ☐ Deployment scheduled (date/time)
- ☐ Post-deployment smoke test plan ready

Notes & Known Issues

Known Limitations

- ☐ Scene 03 rhythm game: if rhythm is difficult for young children, consider adding visual beat indicator
- ☐ Scene 04: if card-slide sometimes jerky on older mobile devices, consider reducing duration

Future Enhancements (Post-Freeze)

- ☐ Symbol Sidebar: add detailed symbol encyclopedia entries
- ☐ Idle Hints: add more granular levels (wobble → glow → VO → sparkle → voice)
- ☐ Gesture System: expand Ganesha pose library with T28 implementation
- ☐ VO: consider pre-recorded MP3s (T14) instead of TTS for production polish

End of Symbol Mountain Production Freeze Checklist

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Prepared for: Ganesha My Bestie Production Freeze Review

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